Reviewer answers: O-RAN S-plane timing security

Evidence status: 2026-11-03 **Primary project baseline:** 168-run classification campaign plus the 5 October 2026, 140-run recovery-loop evaluation. **Reading rule:** MEASURED, CONFIGURED and REFERENCE claims are kept separate. The software testbed measured protocol state, not RF service or nanosecond clock error.

Q1. What kind of attack, and by whom?

The project studies eight PTP timing attacks on the Open Fronthaul synchronization plane: rogue grandmaster (A1), Sync/Follow_Up spoofing (A2), replay (A3), PTP flooding (A5), rogue boundary clock (A8), interception/removal (C1), malformed frames (C2) and whole-second time-property spoofing (C3). The attacker is defined by access and position: an internal attacker or injector that can transmit on the fronthaul, or a man-in-the-middle that can intercept traffic. Realistic devices include a compromised O-RU/O-DU, switch or boundary clock, or an unauthorized host attached to the Ethernet segment. ETSI also names actor categories—insiders, nation states, cyber-criminals, hacktivists, cyber-terrorists and script kiddies—but the experiment tests capabilities, not attribution to a named person.

Evidence: `CITATION_VERIFICATION_2026-11.md` S09-01/S09-02; RFC 7384 §3.1: “Internal attackers either have access to a trusted segment …”; ETSI TR 104 106 §7.2; `00_LATEST_PRESENTED_DECK_AND_DELIVERABLES/ORAN_SPlane_Packets_to_Classification_2026-10-05.xlsx`.

Limits: The attackers are Linux namespaces on one host. The project did not compromise commercial equipment or identify a real threat actor.

Q2. What references support the work?

The main primary references are RFC 7384 for PTP threat classes and security requirements; ETSI TR 104 106 for O-RAN threat agents and T-SPLANE threats; ETSI TS 104 105 for O-RAN security test families; ETSI TS 103 982 and TS 103 859 for architecture and the C/U/S-plane boundary; ITU-T G.8271.1 and G.8275.1 for timing budgets and the telecom PTP profile; and 3GPP TS 38.104 for radio TAE limits. TIMESAFE supplies published experimental impact, while the Armstrong and Ruffini tutorials expose O-RAN/3GPP timing concepts in a concise form. Public copies, URLs, retrieval dates, byte sizes and SHA-256 hashes are recorded locally. IEEE 1588-2019 remains paywalled and is explicitly marked not accessible.

Evidence: `01_CURRENT_SPlane_SelfHealing/literature-survey/papers/Standards/SOURCES_MANIFEST.md`; `CITATION_VERIFICATION_2026-11.md` (63 CONFIRMED, 2 WRONG_CLAUSE, 3 NOT_SUPPORTED, 14 NOT_ACCESSIBLE).

Limits: Clause-level IEEE 1588 claims require licensed access before publication. Secondary tutorials are not substitutes for normative text where safety-critical compliance is asserted.

Q3. Does it only work on O-RAN or another architecture too?

The method is PTP-oriented, not inherently O-RAN-only. Message legality, sequence continuity, delivered-versus-declared rate, identity allow-lists and maintenance context can transfer to other networks that use IEEE 1588, including telecom, power, broadcast/audio and industrial profiles. Profile constants must be re-derived: domain, multicast addresses, message cadence and clock-selection rules differ. This project demonstrated only the ITU-T G.8275.1 full-timing-support profile in an O-RAN-like fronthaul topology.

Evidence: `CITATION_VERIFICATION_2026-11.md` S40-01–S40-03; RFC 7384 scope; ITU-T G.8275.1 §§6.1–6.3; v8 slide 40.

Limits: No non-O-RAN deployment or non-G.8275.1 profile was executed, so portability is an engineering proposition rather than a measured result.

Q4. From which model did we get the KPIs?

No model generates the KPIs. Wire fields come from captured PTP messages, daemon timing values come from `ptp4l` logs, management state comes from `pmc`, and operator intent comes from provisioned `context.json`/state. The rule arm consumes protocol and context evidence; the ML arm consumes six servo-derived features. Thus the model is a consumer of KPIs, not their source.

Evidence: v8 slide 20; `00_LATEST_PRESENTED_DECK_AND_DELIVERABLES/verification_records/V8_SLIDE_SOURCE_MAP.md`; `01_CURRENT_SPlane_SelfHealing/oran_splane_selfhealing/ml_comparison_output/ML_VS_RULE_COMPARISON.md`; linuxptp `pmc.8` (`-d` domain option) confirmed in `CITATION_VERIFICATION_2026-11.md` S20-02.

Limits: Campaign `pmc` queries omitted `-d 24` and returned no data; the 5 October observer corrected this. Servo features remain software-timestamped and do not establish hardware-grade nanosecond accuracy.

Q5. Where did the dataset come from?

The principal dataset was generated by the team on a six-daemon linuxptp 4.0 software testbed using two Linux bridges and network namespaces. The corrected campaign contains 168 runs: 14 scenarios × 12 repetitions, with 96 attack, 60 benign and 12 ambiguous runs. It stores packet-derived tables, context, decisions and RU daemon logs; `EVALUATION_V4.json` is the primary evaluation record. TIMESAFE captures are separate external real-hardware reference data and were not passed off as the project's own campaign.

Evidence: `01_CURRENT_SPlane_SelfHealing/oran_splane_selfhealing/gap_coverage_2026-09-20/corrected_final/EVALUATION_V4.json` (`n_runs=168` and per-scenario counts); `00_LATEST_PRESENTED_DECK_AND_DELIVERABLES/verification_records/SESSION_RECORD_2026-10-03.md` §2 and `V8_SLIDE_SOURCE_MAP.md` (recording the unversioned workspace `dataset/DATASET_MANIFEST.csv`); v8 slides 22–23.

Limits: The campaign is a software testbed on one host, not a production O-RAN network. Several archived synthetic/older datasets are superseded and must not be pooled with this result.

Q6. Who performs these attacks and what are their impacts?

An internal injector can execute A1, A2, A3, A5, C2 or C3 after gaining transmit access; an in-path device can execute C1 or act as the A8 relay. Measured impact is protocol-specific: A1 and C3 captured RU1/RU2 for a median 38.5 seconds in the recovery control arm, while sustained C1 left them unhealthy for 39.0 seconds. A2, A3, A5, A8 and C2 caused 0 seconds of measured primary-RU impact in that topology, although the loop contained their ingress. Published service effects such as cell drop or RU crash are reference evidence, not outcomes measured by this project.

Evidence: `IMPACT_EVIDENCE_SCOPE.md`; `03_RECOVERY_LOOP_S-PLANE/RESULTS_2026-10-05.md` §§1–2; TIMESAFE pp. 12–13; ETSI TR 104 106 T-SPLANE-01…04.

Limits: `free_running 1` prevented physical clock steering. No UE throughput, RF, cell availability or absolute time-error outcome was measured.

Q7. Again, who does these attacks and how?

The experimental actor is a host or network function with the required network position. A1/A8 run additional real `ptp4l` clocks; A2/C2/C3 inject crafted PTP frames; A3 records and replays live frames; A5 sends high-rate bursts; C1 blackholes the boundary clock's downstream port. These are executed packet-path mechanisms, not simulated labels. The corresponding real-world actor would need compromised or unauthorized access to an O-RU/O-DU, a fronthaul switch/clock, or the Ethernet link.

Evidence: v8 slides 10 and 24; campaign archive `splane_campaign_CORRECTED_2026-09-20.tgz` (`run/scenarios.sh`, `run/run_gap.sh`, `run/inject_malformed.py`, `run/inject_wholesecond.py`) and `03_RECOVERY_LOOP_S-PLANE/code/FROZEN_RL.json`; RFC 7384 §3.1.2.

Limits: The testbed does not demonstrate how initial compromise is achieved. It starts after the actor has the stated access.

Q8. Were the eight attacks existing or made up? Where were they referred from?

All eight belong to published threat categories. A1/A8 map to rogue-master behavior, A2/C3 to spoofing, A3 to replay, A5 to time-protocol DoS, C1 to interception/removal, and C2 to exploitation/unexpected-input testing. C3's exact `leap61/currentUtcOffset` combination and C2's exact malformed values are project constructions inside those published categories. ETSI's unexpected-input test does not prescribe the same three C2 mutations, so the deck must not imply an exact reproduction.

Evidence: RFC 7384 §§3.2.2–3.2.5, 3.2.9, 3.2.11; ETSI TR 104 106 T-SPLANE-01…04; ETSI TS 104 105 §§11.1.5 and 24.2.1; `CITATION_VERIFICATION_2026-11.md` S11-01–S11-10.

Limits: IEEE 1588 §8.2.4 was not independently verified because the standard is paywalled.

Q9. What are the citations for the attack references?

A1/A8 cite RFC 7384 §3.2.4 and ETSI T-SPLANE-03; A2/C3 cite RFC 7384 §3.2.2; A3 cites §3.2.3 and TIMESAFE §4.1.2; A5 cites RFC §3.2.9, T-SPLANE-01 and ETSI tests 11.1.5.1.1/24.2.1.1. C1 cites RFC §3.2.5, T-SPLANE-04 and ETSI 11.1.5.3.1. C2 cites RFC §3.2.11 and ETSI 24.2.1.2. The citation audit records a short supporting extract and verdict for every source family used on the relevant deck slides.

Evidence: `CITATION_VERIFICATION_2026-11.md`; locally hashed sources in `papers/Standards/SOURCES_MANIFEST.md`.

Limits: The audit flags 14 inaccessible citations, chiefly IEEE 1588 and unstored secondary sources; those should not be presented as independently confirmed.

Q10. Explain the DU layers and architecture; what attacks occur there and who performs them?

The O-CU-CP hosts RRC and control-plane PDCP; the O-CU-UP hosts SDAP and user-plane PDCP; the O-DU hosts RLC, MAC and High-PHY; and the O-RU hosts Low-PHY and RF. Open Fronthaul split 7-2x connects O-DU to O-RU and carries C, U, S and M planes. This project operates on the S-plane timing path and emulates an LLS-C3 structure, where the O-DU is not in the synchronization chain; it does not emulate O-DU RLC/MAC/High-PHY behavior. The attacks therefore hit PTP processing, parent selection, time properties or message availability at the O-RU/clock path, performed by a transmitter or in-path device with fronthaul access.

Evidence: ETSI TS 103 982 §§3.1, 6.3.6, 6.4.7; ETSI TS 103 859 §11.2.2; `CITATION_VERIFICATION_2026-11.md` S04-01, S05-01, S06-01; v8 slides 4–6.

Limits: No real O-DU, C-plane, U-plane, scheduler, RF chain or 5G core was exercised.

Q11. How should this have been presented better?

The v8 deck already added slide 3 as a reviewer-question map and added architecture, threat-agent, provenance, KPI-source and applicability slides. The `webapp-sim` branch also provides a web simulator presentation path; its existence is a delivery artifact, not evidence that the review was satisfied. This evidence pack improves the presentation by separating measured protocol results from reference service impacts, adding one answer per reviewer question, auditing each citation, and marking superseded statements. A future deck should use the per-attack impact table, place the evidence status beside each number, and replace the slide 6 `T-FRHAUL-01` citation.

Evidence: v8 slide 3; `V8_SLIDE_SOURCE_MAP.md`; `CITATION_VERIFICATION_2026-11.md` errata; Git branch list for `webapp-sim`.

Limits: No claim is made about reviewer satisfaction, and the frozen v8 deck was not edited.

Q12. Which module will these attacks affect?

A1/A8 affect BMCA parent selection and the timing hierarchy; A2/A3 affect Sync/Follow_Up consumption and sequence/timestamp consistency; A5 affects PTP processing capacity; C1 affects message availability and timeout/holdover behavior; C2 affects the PTP parser/stack; C3 affects Announce time-properties and parent selection. In O-RAN terms the target is the Open Fronthaul S-plane termination and timing behavior at the O-RU and intervening clocks. The project did not demonstrate direct effects on RLC, MAC, High-PHY, Low-PHY signal processing, C-plane or U-plane traffic.

Evidence: `IMPACT_EVIDENCE_SCOPE.md`; `ORAN_SPlane_Packets_to_Classification_2026-10-05.xlsx` Source/Destination columns; ETSI TS 103 982 §6.3.6.

Limits: Literature links timing failure to radio service, but this testbed stops at protocol-state evidence.

Q13. Classify the KPIs so recovery can proceed.

Recovery uses six evidence classes. Protocol-legality KPIs find malformed traffic; identity/topology KPIs find unauthorized GM/BC/source identities; continuity/rate KPIs find replay, flooding or removal; time-property KPIs find C3; operator context separates planned from unplanned changes; timing-quality KPIs grade severity. The first four can drive ATTACK/isolation, context can produce BENIGN/tolerate or UNKNOWN/escalate, and timing quality alone does not determine malicious intent. GNSS, SyncE and independent oscillator evidence are unavailable in this software testbed.

Evidence: v8 slide 21; `03_RECOVERY_LOOP_S-PLANE/RECOVERY_LOOP_DESIGN.md`; campaign archive `splane_campaign_CORRECTED_2026-09-20.tgz` (`run/decision_rule.py` and `run/decision_rule_v3.py`).

Limits: The classification is tied to the frozen G.8275.1 feature set. Hardware-only evidence requires a different apparatus.

Q14. What is the source of attack?

“Source” has two meanings and both are now explicit. The network source is the ingress device/port: rogue clock namespace, RU-side injector, two-port relay or blackholed BC port. The knowledge source is the published threat/test family: RFC 7384, ETSI TR 104 106, ETSI TS 104 105 and TIMESAFE. Per-run context and packet identity locate the experimental source; the literature does not identify a real attacker in this project.

Evidence: 2026-10-05 workbook `Source (of attack or fault)` and `Attack devices` columns; v8 slides 10–11; `CITATION_VERIFICATION_2026-11.md`.

Limits: Source localisation is within the two-bridge testbed. It does not prove attribution across a production supply chain or remote compromise path.

Q15. What are the attacks, where and how exactly do they impact?

The eight attacks and exact measured effects are tabulated in `IMPACT_EVIDENCE_SCOPE.md`. A1 and C3 changed the primary RUs' followed source for 38.5 seconds median in control; C1 removed their usable master for 39.0 seconds median. A2, A3, A5, A8 and C2 were observable/containable but produced 0 seconds primary-RU protocol-health impact in the recovery control arm. The loop reduced A1/C3 median unhealthy time to 2.0 seconds and C1 to 2.5 seconds, restoring 5/5 in each case.

Evidence: `03_RECOVERY_LOOP_S-PLANE/RESULTS_2026-10-05.md` §1; `IMPACT_EVIDENCE_SCOPE.md`; raw `results/recovery_eval_runs_r13-r17.tgz`; `EVALUATION_RL.json`.

Limits: These figures measure parent/grandmaster health, not clock offset, time above 1.5 µs, packet-service quality, RF or UE experience.

Q16. Is the approach reactive or proactive?

The implemented loop is reactive: it acts after packet evidence or timing-service loss appears. Preventive provisioning—allow-lists, port roles, maintenance context, a standby BC and frozen rules—makes the reactive decision safer and faster. It is not predictive: no model forecasts an attack before its first observable action. Slow oscillator degradation could be trended in future, but B6 forecasting was not implemented or evaluated.

Evidence: `03_RECOVERY_LOOP_S-PLANE/REACTIVE_VS_PROACTIVE.md`; `RECOVERY_LOOP_DESIGN.md`; `PREREGISTRATION.md`; RFC 7384 §5.9 redundancy quote in `CITATION_VERIFICATION_2026-11.md` D08.

Limits: The evaluation has five repetitions per arm/scenario and one software host; it does not establish production forecasting or general proactive autonomy.

Superseded delivery-status statement

V8 slide 38 states “closed-loop healing not validated” under M5, “None in the 168-run campaign” under Automated corrective action, and “No MTTR, recovery-time, availability or downtime figure was measured on this testbed.” Those statements described the earlier 168-run classification campaign and are superseded by the 5 October recovery-loop evaluation. The later evaluation executed 140 live runs; restored 5/5 for every attack scenario; reduced A1 and C3 median primary-RU unhealthy time from 38.5 s to 2.0 s; reduced sustained C1 from 39.0 s to 2.5 s; and verified 35/35 isolations plus 6/6 standby activations in the artifacts. This is closed-loop protocol-state recovery, not nanosecond time-error recovery or digital-twin rehearsal.

Evidence: `03_RECOVERY_LOOP_S-PLANE/RESULTS_2026-10-05.md` §§1, 2 and 5; `PREREGISTRATION.md`; `code/FROZEN_RL.json`.

Limits: H3 failed because one of 25 benign loop runs caused a disruptive standby action during a genuine B3 timing outage. The sample is small, clock steering remained disabled, and no real O-RAN service was measured.

End of audited answer sheet